PHYS 253: Thermal Physics

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Abstract

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1 Introduction

1.1 Course Information

The textbook is *An Introduction to Thermal Physics* by Daniel V. Schroeder. Grading is divided as follows:

- 10% problem sets (~ 6, submitted as handwritten PDF scans)[1]
- 50% two midterms
- 40% final exam

[1] Only one or two problems on each assignment are marked for submission and grading.

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7 Statistical Mechanics

7.1 Introduction to Probability and Statistics

7.1.1 Microstates, Macrostates and Multiplicity

Definition 1 (Microstate)**.** *A microstate is a specific configuration of the particles in a system.*

Definition 2 (Macrostate)**.** *A macrostate is a description of the system in terms of macroscopic properties, such as temperature, pressure, and volume.*

Definition 3 (Multiplicity)**.** *The multiplicity* Ω *of a macrostate is the number of microstates corresponding to that macrostate.*

The probability of any particular macrostate can be written as

$$P(n) = \frac{\Omega(n)}{\Omega(all)}. \tag{1}$$

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